Instructions

1. **Attempt all questions.**
2. **Write the Python code and also mention the output.**
3. **Use meaningful and proper variable names.**
4. **Use the NumPy library wherever required.**

Question 1

Create a NumPy array of numbers from 1 to 20 and reshape it into a 4×5 matrix.

Expected Output

```
[[ 1  2  3  4  5]
 [ 6  7  8  9 10]
 [11 12 13 14 15]
 [16 17 18 19 20]]
```

Question 2

Create a 5×5 matrix with random integers between 1 and 100 and find:

- Minimum value
- Maximum value

Example Output

```
Matrix:
[[23 45 12 89 34]
 [67 11 90 54 32]
 [76 28 19 47 65]
 [14 73 52 39 81]
 [60 21 48 70 16]]

Min = 11
Max = 90
```

(Values may vary because of random numbers.)

Question 3

Create a NumPy array of numbers from 1 to 10 and replace all even numbers with 0.

Expected Output

[1 0 3 0 5 0 7 0 9 0]

Question 4

Create two 3×3 matrices and perform matrix multiplication.

Matrix A

```
[[1 2 3]
 [4 5 6]
 [7 8 9]]
```

Matrix B

```
[[9 8 7]
 [6 5 4]
 [3 2 1]]
```

Expected Output

```
[[ 30  24  18]
 [ 84  69  54]
 [138 114  90]]
```

Question 5

Create a NumPy array of numbers from 1 to 25 and reshape it into a 5×5 matrix.
Extract the middle 3×3 sub-matrix.

Expected Output

```
[[ 7  8  9]
 [12 13 14]
 [17 18 19]]
```

Question 6

Create a 4×4 matrix and calculate:

- Row-wise sum
- Column-wise sum

Example Matrix

```
[[1 2 3 4]
 [5 6 7 8]
```

```
 [9 10 11 12]
 [13 14 15 16]]
```

Expected Output

```
Row Sum:
[10 26 42 58]

Column Sum:
[28 32 36 40]
```

Question 7

Create a NumPy array of numbers from 1 to 15 and find all numbers greater than 10.

Expected Output

```
[11 12 13 14 15]
```

Question 8

Create a 3×3 matrix and calculate the transpose of the matrix.

Matrix

```
[[1 2 3]
 [4 5 6]
 [7 8 9]]
```

Expected Output

```
[[1 4 7]
 [2 5 8]
 [3 6 9]]
```

Question 9

Generate a NumPy array of 10 random numbers between 0 and 1 and normalize the array between 0 and 1.

Example Output

```
Original Array:
[0.54 0.21 0.87 0.12 0.63 ...]

Normalized Array:
[0.56 0.12 0.95 0.00 0.67 ...]
```

(Values will vary.)

Question 10

Create a NumPy array of numbers from 1 to 10 and reverse the array.

Expected Output

[10 9 8 7 6 5 4 3 2 1]